

Encryption & Decryption

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	17	18	19	20	21	22	23	24	25	26				
P	Q	R	S	T	U	V	W	X	Y	Z				

Formula: Encryption: $CT = E(P+n) \bmod 26$ where n is key
 Decryption: $PT = D(C-n) \bmod 26$
 P- Plain Text
 C- Cipher Text

I. Encryption: Text: CHASE YOUR DREAMS

1. Caesar Cipher

For Caesar cipher, $n=3$ so.

$$CT = E(PT+3) \bmod 26$$

$C = 3+3 = 6 = F$	$Y = 25+3 = 28 \equiv 2 = B$	$D = 4+3 = 7 = G$
$H = 8+3 = 11 = K$	$O = 15+3 = 18 = R$	$P = 16+3 = 19 = U$
$A = 1+3 = 4 = D$	$U = 21+3 = 24 = X$	$E = 5+3 = 8 = H$
$S = 19+3 = 22 = V$	$R = 18+3 = 21 = U$	$A = 1+3 = 4 = D$
$E = 5+3 = 8 = H$		$M = 13+3 = 16 = P$
		$S = 19+3 = 22 = V$

Ans: FKDVH BRXU QUHDPV

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2. Shift Cipher

Given key = 3. So. $n=3$

$$E: C = E(P+n) \bmod 26$$

$C = 3+7 = 10 = J$	$Y = 25+7 = 32 \equiv 6 = F$	$D = 4+7 = 11 = K$
$H = 8+7 = 15 = O$	$O = 15+7 = 22 = V$	$R = 18+7 = 25 = Y$
$A = 1+7 = 8 = H$	$U = 21+7 = 28 \equiv 2 = B$	$E = 5+7 = 12 = L$
$S = 19+7 = 26 = Z$	$R = 18+7 = 25 = Y$	$A = 1+7 = 8 = H$
$E = 5+7 = 12 = L$		$M = 13+7 = 20 = T$
		$S = 19+7 = 26 = Z$

Ans: JOHZL FVBY KYLHTZ

3. ROT13

$$\text{Key} = 13 = n; E: C = E(P+13) \bmod 26$$

$C = 3+13 = 16 = P$	$Y = 25+13 = 38 \equiv 12 = L$	$D = 4+13 = 17 = Q$
$H = 8+13 = 21 = U$	$O = 15+13 = 28 \equiv 2 = B$	$R = 18+13 = 31 \equiv 5 = F$
$A = 1+13 = 14 = N$	$U = 21+13 = 34 \equiv 8 = H$	$E = 5+13 = 18 = R$
$S = 19+13 = 32 \equiv 6 = F$	$R = 18+13 = 31 \equiv 5 = F$	$A = 1+13 = 14 = N$
$E = 5+13 = 18 = R$		$M = 13+13 = 26 \equiv 0 = Z$
		$S = 19+13 = 32 \equiv 6 = F$

Ans: PUNER LBHF QERNZF

II. Decryption

1. Caesar Cipher - WRWD00B VHFUHW PHVVDJM

$$\text{key} = 3 = n \quad P = D(C - 3) \bmod 26$$

$W = 23 - 3 = 20 = T$	$V = 22 - 3 = 19 = S$	$P = 16 - 3 = 13 = M$
$R = 18 - 3 = 15 = O$	$H = 8 - 3 = 5 = E$	$H = 8 - 3 = 5 = E$
$W = 23 - 3 = 20 = T$	$F = 6 - 3 = 3 = C$	$V = 22 - 3 = 19 = S$
$D = 4 - 3 = 1 = A$	$U = 21 - 3 = 18 = R$	$V = 22 - 3 = 19 = S$
$O = 15 - 3 = 12 = L$	$H = 8 - 3 = 5 = E$	$D = 4 - 3 = 1 = A$
$O = 15 - 3 = 12 = L$	$W = 23 - 3 = 20 = T$	$J = 10 - 3 = 7 = G$
$B = 2 - 3 = 25 = Y$		$H = 8 - 3 = 5 = E$

Ans: TOTALLY SECRET MESSAGE

2. Shift Cipher - SJT IBTW YNRJ

$$\text{key} = n = 5 \quad P = D(C - 5) \bmod 26$$

$S = 19 - 5 = 14 = N$	$I = 9 - 5 = 4 = D$	$Y = 25 - 5 = 20 = T$
$J = 10 - 5 = 5 = E$	$B = 2 - 5 = 23 = W$	$N = 14 - 5 = 9 = I$
$T = 9 - 5 = 4 = D$	$T = 20 - 5 = 15 = O$	$R = 18 - 5 = 13 = M$
	$W = 23 - 5 = 18 = R$	$J = 10 - 5 = 5 = E$

Ans: NED DWOR TIME

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3. ROT13 - FRAQ ZBER PBSSPR

$$\text{key} = n = 13 \quad P = D(C - 13) \bmod 26$$

$F = 6 - 13 = 19 = S$	$Z = 26 - 13 = 13 = M$	$P = 16 - 13 = 3 = C$
$R = 18 - 13 = 5 = E$	$B = 2 - 13 = 15 = O$	$B = 2 - 13 = 15 = O$
$A = 1 - 13 = 14 = N$	$E = 5 - 13 = 18 = R$	$S = 19 - 13 = 6 = F$
$Q = 17 - 13 = 4 = D$	$R = 18 - 13 = 5 = E$	$S = 19 - 13 = 6 = F$
		$R = 18 - 13 = 5 = E$
Ans: SEND MORE COFFEE		$R = 18 - 13 = 5 = E$